

Homework 2

Give exact answers and justify your reasoning. Do not use electronic tools. Use the limit definition, uniqueness, and the sum law covered in Lecture 3; do not assume other limit laws or continuity theorems. All functions are real-valued. Unless otherwise specified, a function whose limit at a is considered is defined on an open interval containing a , except possibly at a itself. Every condition on x refers to points in the domain of the function or the common domain of the functions involved.

1. Write the epsilon-delta definition of $\lim_{x \rightarrow -1} f(x) = 2$, and then write its negation.

2. Let $m, b, a \in \mathbb{R}$. Prove directly from the definition that

$$\lim_{x \rightarrow a} (mx + b) = ma + b.$$

Your proof must include the case $m = 0$.

3. Prove directly from the definition that $\lim_{x \rightarrow -1} x^2 = 1$. First find a constant bound for $|x - 1|$ under the restriction $|x + 1| < 1$.
4. Prove directly from the definition that $\lim_{x \rightarrow 0} x^2 = 0$. Give a choice of δ that does not require a separate restriction such as $\delta \leq 1$.

5. Define

$$f(x) = \begin{cases} 3x + 1, & 0 < |x - 1| < 1/2, \\ 100, & \text{otherwise.} \end{cases}$$

Find $\lim_{x \rightarrow 1} f(x)$ and prove your answer directly from the definition. What is $f(1)$?

6. Construct a function on $\mathbb{R} \setminus \{0\}$ whose limit at zero is 2, but whose value is never 2. Prove that your function has these properties.

7. Suppose both

$$\lim_{x \rightarrow 1} f(x) = 2c - 1 \quad \text{and} \quad \lim_{x \rightarrow 1} f(x) = 5 - c$$

hold. Determine c and the common limit. State the theorem you use.

8. Suppose that whenever $0 < |x - 1| < 1$,

$$|f(x) - 2| \leq 4|x - 1|, \quad |g(x) + 3| \leq \sqrt{|x - 1|}.$$

Prove $\lim_{x \rightarrow 1} (f(x) + g(x)) = -1$ by giving an explicit choice of δ for each $\varepsilon > 0$.

9. Suppose $\lim_{x \rightarrow a} f(x) = 2$, $\lim_{x \rightarrow a} g(x) = -1$, and $\lim_{x \rightarrow a} h(x) = 4$. Find the following limits using the sum law and the limit of a constant:

$$(a) \lim_{x \rightarrow a} (f(x) + g(x) + h(x) + 5), \quad (b) \lim_{x \rightarrow a} (f(x) + f(x) + g(x)).$$

10. Use the negation of the limit definition to show that $\lim_{x \rightarrow 0} x^2 = 1$ is false. Specify a fixed $\varepsilon_0 > 0$ and, for every $\delta > 0$, a suitable point x .